



AIR UNIVERSITY
DEPARTMENT OF MECHATRONICS ENGINEERING
SEMESTER: SPRING 2026
Computer Programming CE-112

Assignment-3

Instructor: Umer Farooq

Due Date: 7th May 2026

Soft Copy Time: 10:00 PM

Cover Page must be filled as per attached.

Name Ahmad Bilal **Name in ASCII(HEX)** _____

Rollno 2501556 **Rollno:in 4bit Hex** _____

Nameing Convention of file should be done as descibed in file format to avoid marks deduction

Course_Serial_Section __No_240_XXX__Name_A3_15_NOV__V1

Example= 10_F24A_21841_Haris_A3_7_OCT_V1

Question No	Write Status(Complete/Not complete and issue that you faced (compulsory Question will be checked on bases of this index)	Remarks Must be filled for every question	
1			
2			
3			
4			
5			
N	Add more Rows when required		

	according to number of questions attempted		
--	--	--	--

Total Questions are 5 covering your Project, add all the following

For all questions you must do the following procedures

- Make block diagram for all function inputs & outputs
- Make Flowchart for code
- Write code in C++
- Comment code
- Show all calculation
- Copying other code will get you 0 marks
- Screen Shots of Code along with simulations are to be attached h)

All program should be made using functions

Design the loop below and show the calculations. Also, explain everything in a video and upload

- Write and test the following function that returns through its reference parameters both the maximum and the minimum values stored in an array: void getExtremes(float& min, float& max, float a[], int n);
- Write and test the following function: double stdev(double x[], int n);

The function returns the standard deviation
of a data set of n numbers $x_0, \dots, x_{n-1}$

- Write a and test a program to calculate write a program to multiply two 3x3 matrices .Make with function that can pass array to each function.

$$s = \sqrt{\frac{\sum_{i=0}^{n-1} (x_i - \bar{x})^2}{n - 1}}$$

GetValue(PassArray)
ComputeAxA(PassArray)
PrintArray(PassArray)

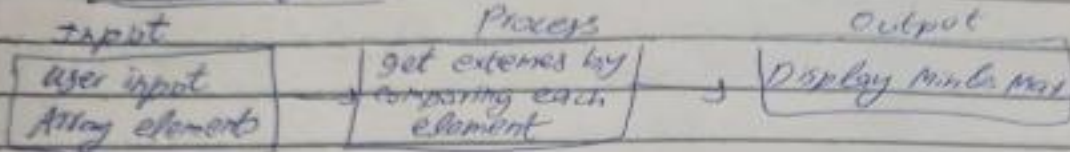
4. Write a program that takes a string from the user. The program then calculates the
 - ✓ Total number of Characters
 - ✓ Total number of Spaces
 - ✓ Total number of Tabs
 - ✓ Total number of Lines
 - ✓
5. Write a program to calculate **mean**, **median** and **mode** of an array using function

CP

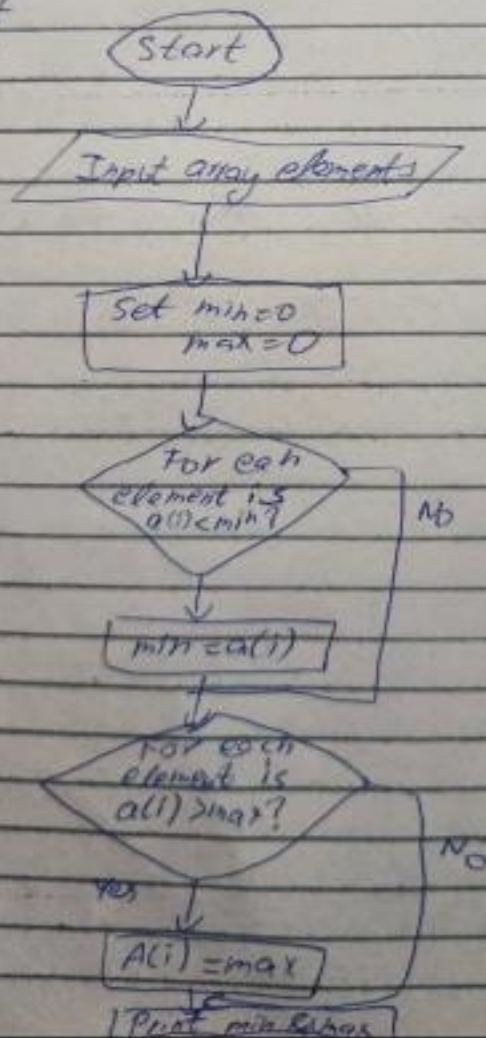
A2

(CODE #01)

=) Block diagram.



=) Flowchart



CODE:-

```
#include <iostream>
using namespace std;
void get Extremes (float &min, float &max, float a[], int n)
{
    min = a[0];
    max = a[0];
    for (int i=1; i<n; i++)
    {
        if (a[i] < min)
            min = a[i];
        if (a[i] > max)
            max = a[i];
    }
}

int main() {
    float arr[5];
    float min, max;
    cout << "Enter 5 elements: \n";
    for (int i=0; i<5; i++)
        cin >> arr[i];
    get Extremes (min, max, arr, 5);
    cout << "Minimum=" << min << endl;
    cout << "Maximum=" << max << endl;
    return 0; }
```

=> Dry run (output):

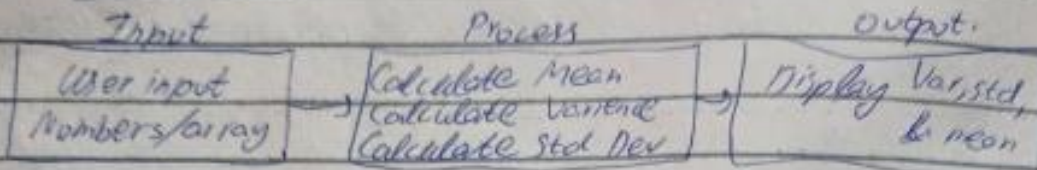
If array = {10, 5, 25, 40, 18}

Min = 5

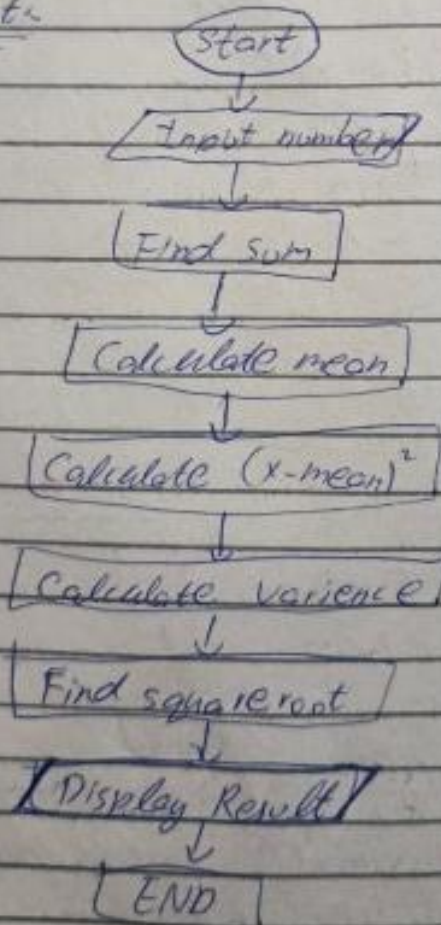
max = 40

← (CODE # 02) →

=> Block diagram



=> Flowchart



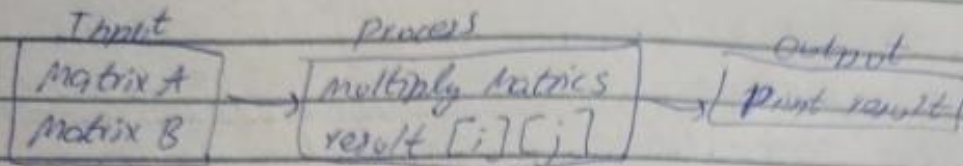
CODE

```
#include <iostream>
#include <math>
using namespace std;
double stdev (double x[], int n)
{
    double sum = 0, mean, Variance = 0;
    for (int i = 0; i < n; i++)
        sum += x[i];
    mean = sum / n;
    for (int i = 0; i < n; i++)
        Variance = Variance + (x[i] - mean) * (x[i] - mean);
    return sqrt(Variance / n);
}

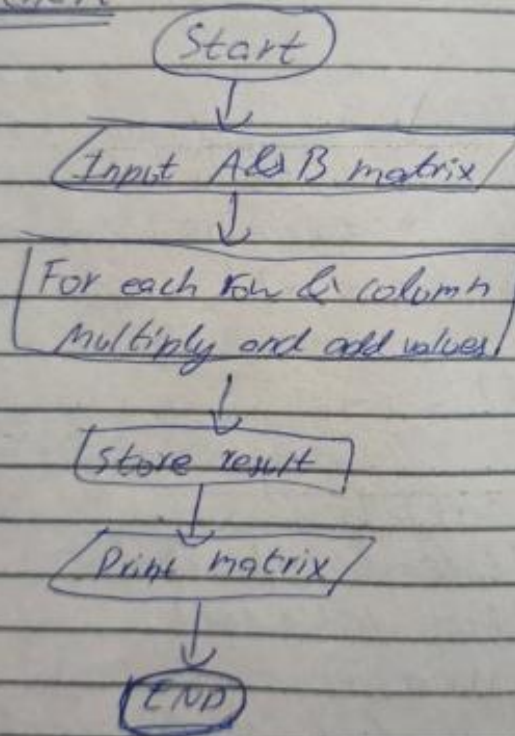
int main()
{
    double arr[5];
    cout << "Enter 5 numbers: \n";
    for (int i = 0; i < 5; i++)
        cin >> arr[i];
    cout << "Standard Deviation = " << stdev(arr, 5);
    return 0;
}
```

(== (CODE #03) ==)

=> Block diagram:-



=> Flowchart



CODE:-

```
#include <iostream>
using namespace std;
void Get Value (int arr [3][3])
{
```



```

for (int i=0; i<3; i++)
    for (int j=0; j<3; j++)
        cin >> arr[i][j];
}

void compute AxA (int a[3][3], int b[3][3], int res
[3][3]) {
    for (int i=0; i<3; i++)
    {
        for (int j=0; j<3; j++)
        {
            result[i][j] = 0;
            for (int k=0; k<3; k++)
                result[i][j] += a[i][k] * b[k][j];
        }
    }
}

void Print Array (int arr[3][3])
{
    for (int i=0; i<3; i++)
    {
        for (int j=0; j<3; j++)
            cout << arr[i][j] << " ";
        cout << endl;
    }
}

int main ()
{
    int a[3][3], b[3][3], result[3][3];
    cout << "Enter first matrix: \n";
    Get value (a);

    cout << "Enter second matrix: \n";
    Get value (b);
}

```

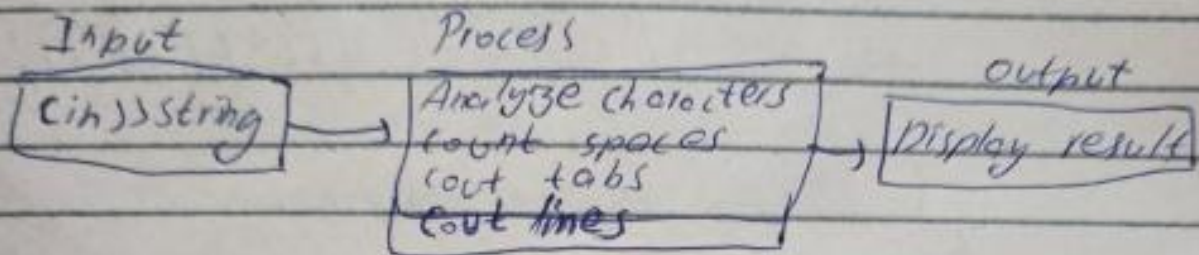
```

compute AxA (a, b, result);
cout << "Result Matrix: \n";
Print Array (result);
return 0;
}

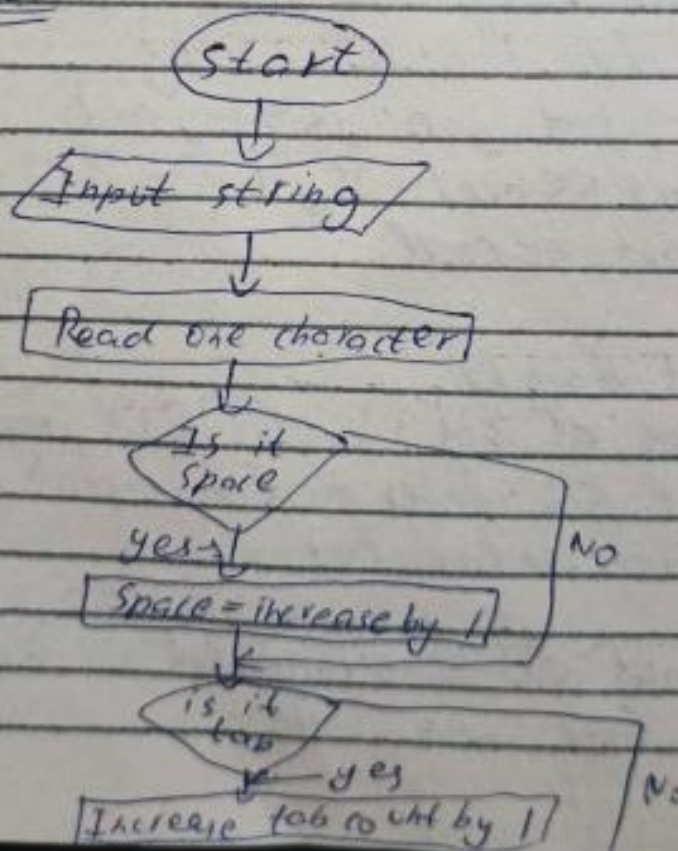
```

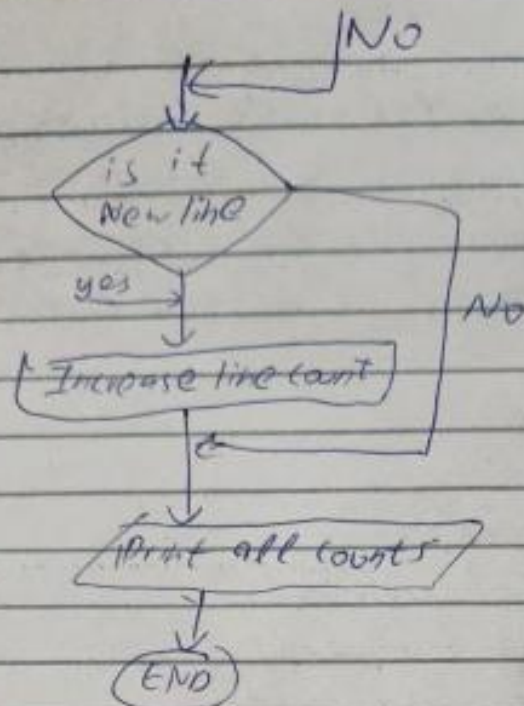
← (CODE #04) →

⇒ Block diagram



⇒ Flowchart





=> CODE :

```

#include <iostream>
using namespace std;
void analyze string (string text)
{
    int chars=0, spaces=0, tabs=0, lines=1;
    for (int i=0; i<text.length(); i++)
    {
        chars++;
        if (text[i] == ' ')
            spaces++;
        if (text[i] == '\t')
            tabs++;
        if (text[i] == '\n')
            lines++;
    }
}
  
```



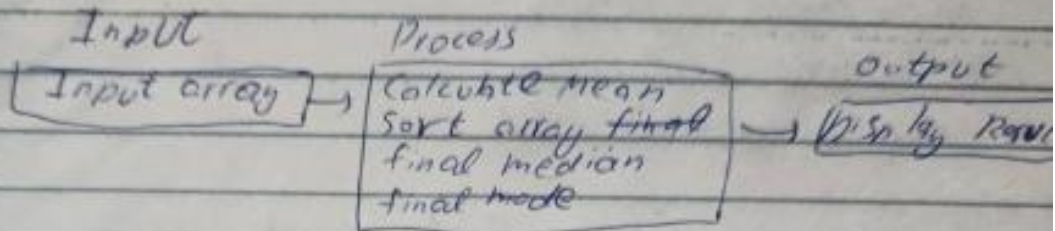
```
cout << "Characters = " << chars << endl;  
cout << "Spaces = " << spaces << endl;  
cout << "Tabs = " << tabs << endl;  
cout << "Lines = " << lines << endl;  
}
```

```
int main()  
{
```

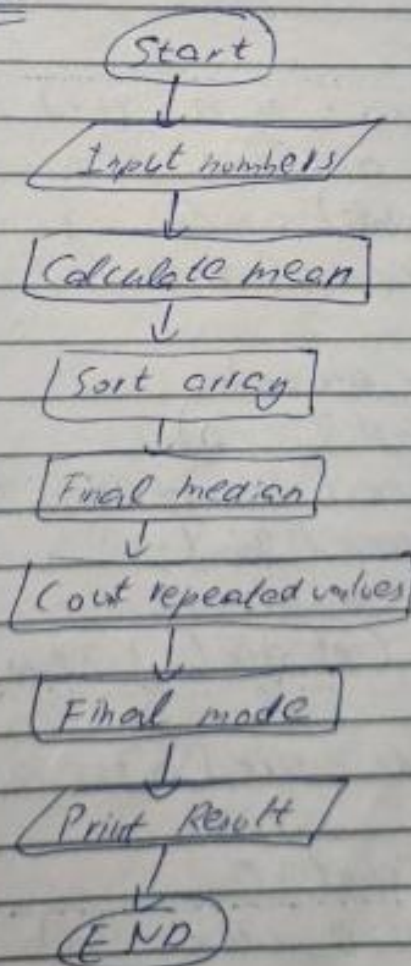
```
    string text;  
    cout << "Enter text: ";  
    get_line (cin, text);  
    analyze_string (text);  
    return 0;  
}
```


==> (CODE #05) <==

=> Block diagram



=> Flow chart



=> Cont:

```
#include <iostream>
```

```
#include <algorithm>
```

```
using namespace std;
```

```
double mean (int arr[], int n)
```

```
{
```

```
    int sum = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
        sum += arr[i];
```

```
    return (double) sum/n; }
```

```
double median (int arr[], int n)
```

```
{
```

```
    sort (arr, arr+n)
```

```
    if (n%2 == 0)
```

```
        return (arr[n/2] + arr[(n/2)-1])/2.0;
```

```
    return arr[n/2];
```

```
}
```

```
int mode (int arr[], int n)
```

```
{
```

```
    int mode = arr[0];
```

```
    int maxCount = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        int count = 1;
```

```
        for (int j = i+1; j < n; j++)
```



```

    { if [arr[i] == arr[j]]
      count++;
    }
    if (count > maxCount)
    {
      maxCount = count;
      mode = arr[i];
    }
  }
}

int main()
{
  int arr[5];
  cout << "Enter 5 number: \n";
  for (int i=0; i<5; i++)
    cin >> arr[i];
  cout << "Mean = " << mean(arr, 5) << endl;
  cout << "Median = " << median(arr, 5) << endl;
  cout << "Mode = " << mode(arr, 5) << endl;
  return 0;
}
return 0;
}

```

